Mohaimin Al Barat

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Education

Virginia Tech — VA, USA
Ph.D. in Computer Science — *Aug. 2022 – Expected Summer 2027*

Khulna University of Engineering & Technology (KUET) — Bangladesh
B.Sc. in Computer Science and Engineering — *2020*

Experience

Graduate Research Assistant — 2022 – Present
Virginia Tech — *VA, USA*

- Designing privacy-enhancing technologies with zero-knowledge proofs, including SpexPay, a privacy-preserving pay-as-you-go payment system for dynamic spectrum sharing (ESORICS 2026)
- Working on anonymous payment solutions, including anonymous yet compliant fiat-to-cryptocurrency exchanges (IEEE INFOCOM 2026)
- Researching blockchain security and scalability; authored a systematization-of-knowledge paper on public blockchain sharding (IEEE ICBC 2024)
- Developing ML-based security for automotive vehicles, including temporal misalignment attacks on and defenses for multimodal sensor fusion in autonomous driving (IEEE SaTML 2026, ICLR 2026), Sybil-resistant V2X pseudonyms (ACM WiSec 2026), and Privacy-preserving EV charging solution with anonymous payment (under submission)
- Building agentic AI solutions for wireless spectrum allocation (under submission)

Application & Software Engineer — 2020 – 2022
Huawei Bangladesh — *Bangladesh*

- Supported operation and maintenance of application systems for bKash Limited, the largest fintech company in Bangladesh

Selected Publications

Full list on Google Scholar.

- **M. M. Al Barat**, H. Yu, S. Li, Y. Xiao, Y. Shi, E. W. Burger, Y. T. Hou, W. Lou. *SpexPay: A Privacy-Preserving Pay-As-You-Go System for Dynamic Spectrum Sharing.* **ESORICS 2026**.
- **M. M. Al Barat**, S. Li, C. Du, Y. T. Hou, W. Lou. *SoK: Public Blockchain Sharding.* **IEEE ICBC 2024**.
- **M. M. Al Barat**, C. Zhang, H. Yu, K. Zeng, Y. T. Hou, W. Lou. *Enabling Emergency Communication via Semantic Radar-Centric ISAC.* **MILCOM 2026**.
- M. H. Shahriar, **M. M. Al Barat**, H. Sundar, N. Zhang, N. Ramakrishnan, T. Hou, W. Lou. *Detecting Temporal Misalignment Attacks in Multimodal Fusion for Autonomous Driving.* **ICLR 2026**.
- M. H. Shahriar, **M. M. Al Barat**, H. Sundar, N. Zhang, N. Ramakrishnan, Y. T. Hou, W. Lou. *Temporal Misalignment Attacks against Multimodal Perception in Autonomous Driving.* **IEEE SaTML 2026**.
- H. Yu, **M. M. Al Barat**, S. Li, M. H. Shahriar, Y. Xiao, P. Papadimitratos, Y. T. Hou, W. Lou. *V-PASS: Sybil-Resistant Pseudonym Self-Provisioning for V2X.* **ACM WiSec 2026**.
- H. Yu, **M. M. Al Barat**, Y. Xiao, Y. T. Hou, W. Lou. *Closing the Visibility Gap: A Monitoring Framework for Verifiable Open RAN Operations.* **IEEE CNS 2025**.

Technical Skills

Languages: C++, Java, Python, Rust, Solidity
Tools: Git, Docker, Linux, Foundry, Wireshark, bbs_plus (BBS+ signatures)
Research Areas: Zero-Knowledge Proofs & Privacy-Enhancing Technologies, Blockchain Security & Scalability, ML Security for Autonomous Vehicles, Agentic AI for Spectrum Allocation